MILITARY SPECIFICATION

CAPSULE ASSEMBLY, TELEMETRY: ARRT-13

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers one type of telemeter, designated Capsule Assembly, Telemetry: ARRT-13, which is used in the Pershing warhead section subassembly during the Pershing flight test program.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents.- The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.
MS 33537 - Insert, Screw Thread, Helical Coil, Coarse and Fine Thread, Standard Dimensions for.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9212881 - TM Structure Assembly, ARRT-13.
9216732 - Capsule Assembly, Telemetry: ARRT-13.
9216733 - TM Insert Assembly, ARRT-13.
9216739 - Plate, Mounting, Lower.
9268476 - Housing Assembly.
9268496 - Cover (Wire and Tac).
9268576 - Container, Packing, Capsule Assembly, Telemetry.
9327977 - Functional Test Procedure for the ARRT-13 Telemeter.

Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Materials and components.- Materials shall conform to applicable specifications and drawings. Each component shall conform to all dimensions and tests specified on the component drawing. The contractor shall have available verifiable proof (i.e., objective evidence (see 6.2)) that materials and components were fabricated, inspected, and tested under controllable conditions as set forth in the contractor's quality control or inspection procedures.

3.2 Assembly.- The ARRT-13 capsule assembly shall comply with all requirements specified on drawing (dwg.) 9216732, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawings and herein.

3.3 First article.- First article inspection is not applicable for the ARRT-13 capsule assembly.

3.4 Continuity.- The insert assembly shall exhibit continuity as measured between terminals listed in Table I, and the resistance shall be a maximum of 0.1 ohm. This test shall be performed prior to conformal coating/connector sealing of the insert assembly. Testing shall be as specified in 4.3.1.

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TABLE I

From	To
J102-C	TMP-5-C
J102-$\overline{C}$	TMP-5-D
J1-C	J1-E
J2-R	J2-S
J2-E	J2-G
J2-E	J2-L

3.5 Functional test.- The insert assembly shall be subjected to, and shall comply with, all applicable requirements of the functional test as specified on dwg. 9327977. This test shall be performed before and after conformal coating/connector sealing of the insert assembly (see sequence of testing in 4.2.3). Testing shall be as specified in 4.3.2.

3.6 Vibration.- The insert assembly shall withstand sinusoidal vibrations as specified in Table II without evidence of deterioration or damage that may impair its intended operation. This test shall be performed prior to conformal coating/connector sealing of the insert assembly (see sequence of testing in 4.2.3). During this test, the insert assembly shall comply with the vibration portion of the functional test procedure (dwg. 9327977). Testing shall be as specified in 4.3.3.

TABLE II

Frequency range (hertz)	Peak g level (± 10% tol.)	Sweep time (minimum)
20-2000-20	2 g	2 minutes
20	5 g	5 seconds

3.7 Serialization.- A serial number shall be assigned by the contractor to each capsule assembly produced. In no event, shall serial numbers be duplicated. Six-letter serial codes shall not be used on the capsule assembly.

3.8 Screw-lock inserts.- Screw-lock inserts shall be installed in accordance with MS 33537.

3.9 Soldering.- All soldering shall be in accordance with MIL-STD-1460.

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3.10 Workmanship.- All parts and accessories shall be constructed, assembled, and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness, marking, cleaning, and freedom of the parts from burrs and sharp edges which may affect performance or result in injury during handling. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Inspection lot.- A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot, and a batch shall mean an inspection batch for the purposes of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

4.2.2 Product inspection examination.- A sample shall be selected at random from each inspection lot in accordance with the major defect or the minor defect table as applicable (see Table III). Any defect found in the sample, selected for the major or minor classification, shall reject the lot. Critical defects shall require 100 percent inspection of all items in the inspection lot. If any critical defect is found in an item, the item shall be removed from the lot. The sequence of inspections for the classification of defects need not be followed so long as all the defect inspections are performed.

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TABLE III. Inspection lot sampling.

Major Defect Table		Minor Defect Table	
Lot Size	Sample Size	Lot Size	Sample Size
1-30	all	1-30	all
31-45	29	31-45	27
46-60	35	46-60	30
61-75	38	61-75	33
76-95	42	76-95	35
96-130	46	96-130	37
131-180	50	131-180	39
181-250	52	181-250	41
251-400	54	251-400	42
401 and over	56	401 and over	44

4.2.2.1 Classification of defects.- Critical, major, and minor defects are defined in MIL-STD-105. Packing inspections (4.2.2.1.6 and 4.2.2.1.7) shall be conducted after the capsule assembly has successfully completed the tests listed in Table IV.

4.2.2.1.1 Plate, Mounting, Lower (see dwg. 9216739 covering a detail of dwg. 9212881).

Categories	Defects	Method of Inspection

Critical: None defined.

Major:

101. Location of countersunk hole in flange surface (4 holes).....Gage
102. Diameter of countersunk hole in flange surface, min. (4 holes).....Gage
103. Countersink of hole, in flange surface, missing or improper (4 places).....Visual
104. Location of "S" hole.....Visual

Minor:

201. Marking missing, improper, or illegible.....Visual
202. Protective finish missing or improper.....Visual
203. Evidence of poor workmanship (see 3.10).....Visual

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4.2.2.1.2 Cover (Wire and Tac)(see dwg. 9268496).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Location of counterbored hole (12 holes)	Gage
	102. Location of drill hole (12 holes)	Gage
	103. Diameter of drill hole, min. (12 holes)	Gage
	104. Location of connector hole (7 holes)	Gage
	105. Location of tapped hole (3 holes)	Gage
	106. Pitch diameter of tapped hole (3 holes)	Gage
	107. Minor diameter of tapped hole, max. (3 holes)	Gage
	108. Location of "Z" mark	Visual
Minor:		
	201. Thickness, min	Gage
	202. Flatness (where specified)	Gage
	203. Marking missing, improper, or illegible	Visual
	204. Protective finish missing or improper	Visual
	205. Evidence of poor workmanship (see 3.10)	Visual

4.2.2.1.3 Housing Assembly (see dwg. 9268476).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Location of small drill hole in flange (12 holes)	Gage
	102. Diameter of small drill hole in flange, min. (12 holes)	Gage
	103. Location of large drill hole in flange (12 holes)	Gage
	104. Diameter of large drill hole in flange, min. (12 holes)	Gage
	105. Location of "S" hole in flange	Gage

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106.	Diameter of "S" hole in flange, min	Gage
107.	Location of "T" hole in flange	Gage
108.	Diameter of "T" hole in flange, min	Gage
109.	Location of large insert hole in top surface (see right side view)(12 holes)	Gage
110.	Location of small insert hole in top surface (see right side view)(12 holes)	Gage
111.	Location of insert hole on rim inside flange area (4 holes)	Gage
112.	Larger diameter of circular groove in flange, min	Gage
113.	Width of circular groove in flange, min	Gage
114.	True position of larger circular groove diameter, in flange, with flange diameter	Gage
115.	Wall thickness, min	Gage
116.	Screw-lock insert missing or improperly installed (28 places) (see 3.8)	Visual
117.	Location of "S," "T," and "W" mark	Visual

Minor:

201.	Length from base of flange to insert hole surface	Gage
202.	Depth of circular groove in flange, min	Gage
203.	Location of cap screw hole on rim inside flange area (inspect prior to assembling of cap screw)(2 holes)	Gage
204.	Flatness of flange surface	Gage
205.	Socket head cap screw on rim, inside flange area, missing or improper (2 screws)	Visual
206.	Marking missing, improper, or illegible	Visual
207.	Protective finish missing or improper	Visual
208.	Evidence of poor workmanship (see 3.10)	Visual

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4.2.2.1.4 TM Insert Assembly, ARRT-13 (see dwg. 9216733).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Conformal coating/connector sealing missing or improper (where applicable)(inspect after conformal coating/connector sealing operation (see sequence of testing in 4.2.3))	Visual
	102. Orientation of terminal boards and components, with upper and lower mounting plates, improper	Visual
	103. Soldering improper (inspect during assembly)(see 3.9)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.2.2.1.5 Capsule Assembly, Telemetry: ARRT-13 (see dwg. 9216732).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Center of gravity	Gage
	102. Weight	Gage
	103. Ability of cable connector to engage and disengage with mating receptacle (7 connectors on cover; 1 connector on battery box)(inspect connector on battery box prior to assembly)	Manual/Gage
	104. Flat head screw, in flange area of housing assembly, missing, loose, or improper (4 screws)	Visual/Manual
	105. Socket head cap screw and washer, securing cover, missing, loose, or improper (12 places)(inspect during assembly)	Visual/Manual

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106. Battery box and components missing, damaged, or improperly assembled (inspect during assembly).....Visual

107. Connector on cover (7 connectors) and connector on battery box damaged (including missing or damaged connector terminals and cut or damaged insert material).....Visual

Minor:

201. Hex head cap screw on cover missing or improper (3 screws).....Visual

202. Identification plate missing, improper, loose, or improperly located.....Visual

203. Marking missing, illegible, or improper.....Visual

204. Evidence of poor workmanship (see 3.10).....Visual

4.2.2.1.6 Container, Packing, Capsule Assembly, Telemetry (prior to securing cover)(see dwg. 9268576).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Screw, securing capsule assembly to dust cover, missing or improperly assembled (8 places)(inspect during assembly).....	Visual/Manual
	102. Any cushioning pad missing or improperly assembled.....	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.10).....	Visual

4.2.2.1.7 Container, Packing, Capsule Assembly, Telemetry (see dwg. 9268576).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Drum improperly sealed (including car seal).....	Visual/Manual

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102. Drum damaged (including punctures, dents, and gouges).....Visual

Minor:

201. Protective finish missing or improper.....Visual
202. Marking missing, improper, or illegible.....Visual
203. Evidence of poor workmanship (see 3.10).....Visual

4.2.3 Inspection testing.- The contractor shall perform all of the tests specified in Table IV. The tests shall be performed in the listed sequence. None of these tests is considered destructive, and samples so tested shall be returned to the lot. Note: Tests 1, 2, and 3 of Table IV shall be performed prior to conformal coating/connector sealing operations of the insert assembly.

TABLE IV. Quality conformance tests.

Test	Test classification
1. Continuity	Major
2. Functional test	Major
3. Vibration	Major
4. Conformal coating/ connector sealing	Major
5. Functional test	Major

4.2.3.1 Continuity (see 3.4), functional test (see 3.5), and vibration (see 3.6).- Each insert assembly shall be subjected to these tests (see applicable paragraph in 4.3). If any insert assembly fails to comply with the applicable requirements, it shall be classified defective and be removed from the lot.

4.2.3.2 Conformal coating/connector sealing (see dwg. 9216733).- The insert assemblies shall be sampled and visually inspected for conformal coating/connector sealing as specified in 4.2 through 4.2.2.1 and 4.2.2.1.4 (defect #101).

4.2.4 Acceptance inspection equipment.

4.2.4.1 Major defects and tests.- Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration procedures designed or spec-

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ified by the contractor (see 6.1.2). The contractor shall obtain approval of such equipment designs, operating instructions, and calibration procedures prior to use on the contract (see 6.3).

4.2.4.2 Minor defects and tests.- The acceptance inspection equipment, operating instructions, and calibration procedures used by the contractor for minor classification inspection shall be approved by the Government representative responsible for acceptance inspection.

4.3 Methods of inspection.- The tolerances specified in this specification are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. Unless otherwise specified, the tests shall be conducted at room temperature (77° ±.10°F).

4.3.1 Continuity.- The continuity of the insert assembly shall be measured by using a Government-approved continuity tester. The insert assembly shall comply with the requirements of 3.4 during this test.

4.3.2 Functional test.- The insert assembly shall be tested on a Government-approved test stand, and unless otherwise specified, the insert assembly shall be tested in the horizontal position (its rotational position is of no importance). The insert assembly shall comply with the requirements of 3.5 during this test.

4.3.3 Vibration.- The insert assembly shall be mounted to a vibration machine with the longitudinal axis in the vertical position (the terminal boards shall be uppermost). The insert assembly shall be subjected to the vibrations specified in Table II along the vertical axis, and it shall comply with the requirements of 3.6 during this test.

5. PACKAGING

5.1 Preservation, packing, and marking.- Preservation, packing, and marking shall be in accordance with dwg. 9268576.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

6.1.1 Procurement requirements.- Title, number, and date of this specification.

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6.1.2 Contract data requirements.- Contractor engineering documentation specified in 4.2.4.1 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated into the contract.

6.2 Objective evidence.- Records of the contractor's quality control and inspections which can be verified.

6.3 Acceptance inspection equipment.- The contractor shall obtain approval of equipment designs, operating instructions, and calibration procedures from Commander, US Army Armament Research and Development Command, Attn: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Drawings and publications.- Drawings and publications listed in Section 2 of this specification under the heading of US Army Armament Research and Development Command (ARRADCOM) may also include drawings and publications prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings and publications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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